```
    worker_opt.zero_grad()
    for i in range(params['epochs']):
        worker_opt.zero_grad()
        values, logprobs, rewards =
run_episode(worker_env,worker_model)  2
        actor_loss,critic_loss,eplen =
update_params(worker_opt,values,logprobs,rewards)
3
        counter.value = counter.value + 1
4
```

- ***1*** Each process runs its own isolated environment and optimizer but shares the model.
- ***2*** The run_episode function plays an episode of the game, collecting data along the way.
- ***3*** We use the collected data from run_episode to run one parameter update step.
- ***4*** counter is a globally shared counter between all the running processes.

The `worker` function is the function that each individual process will run separately. Each worker (i.e., process) will create its own CartPole environment and its own optimizer but will share the actor-critic model, which is passed in as an argument to the function. Since the model is shared, whenever a worker updates the model parameters, they are updated for all the workers. This is shown at a high-level in figure 5.12.

Figure 5.12. Within each process, an episode of the game is run using the shared model. The loss is computed within each process, but the optimizer acts to update the shared actor-critic model that is used by each process.